

DEVELOPMENT OF NEW MODEL FISHERIES FOR SIMULATING LONGLINE CATCH DATA WITH LLSIM

C. Phillip Goodyear¹

SUMMARY

Two idealized fishing fleets were developed to augment LLSIM simulations of complex longline data. They were patterned after Brazilian and Japanese data in the ICCAT Task ii database. These data provided the numbers of hooks fished in 5° latitude-longitude squares for 1958-2018. Each fleet had shallow, intermediate, or deeply-fishing gears. These simulated gears were assigned to individual sets based on previously-reported fishing patterns or the species composition of fleet catches in the ICCAT database. Fishing depths for each gear were adopted from an earlier study of the US fishery. The number of sets in each square were estimated from total hooks using the number of hooks per set for the gear. The sets for each fishery in each 5° square were randomly assigned to habitable 1° LLSIM cells (no land). The resulting file provided the monthly number of sets for each fleet by gear within each 1° square from 1958 to 2018. When used in combination with Swordfish and Blue Marlin species-distribution-model inputs, these fisheries will allow investigators to simulate catches with three-fishery, two-species runs of the LLSIM software.

RÉSUMÉ

Deux flottilles de pêche idéalisées ont été développées pour augmenter les simulations LLSIM des données palangrières complexes. Elles ont été calquées sur les données brésiliennes et japonaises de la base de données de la tâche 2 de l'ICCAT. Ces données ont fourni le nombre d'hameçons pêchés dans des carrés de 5° de latitude-longitude pour la période 1958-2018. Chaque flottille disposait d'engins de pêche pour pêcher en eaux peu profondes, intermédiaires ou profondes. Ces engins simulés ont été assignés à des calées individuelles sur la base des schémas de pêche précédemment déclarés ou de la composition par espèce des captures de la flottille figurant dans la base de données de l'ICCAT. Les profondeurs de pêche de chaque engin ont été adoptées à partir d'une étude antérieure de la pêche des Etats-Unis. Le nombre de calées dans chaque carré a été estimé à partir du nombre total d'hameçons en utilisant le nombre d'hameçons par calée pour l'engin. Les calées de chaque pêcherie dans chaque carré de 5° ont été assignées aléatoirement à des cellules LLSIM de 1° (sans terre). Le fichier résultant fournissait le nombre mensuel de calées pour chaque flottille par engin dans chaque carré de 1° de 1958 à 2018. Lorsqu'elles sont utilisées en combinaison avec les entrées du modèle de distribution des espèces d'espadon et de makaira bleu, ces pêcheries permettront aux enquêteurs de simuler les captures avec des scénarios du logiciel LLSIM avec trois pêcheries et deux espèces.

RESUMEN

Se desarrollaron dos flotas pesqueras idealizadas para aumentar las simulaciones LLSIM de los datos complejos de palangre. Se diseñaron según los datos brasileños y japoneses de la base de datos de Tarea 2 de ICCAT. Estos datos proporcionaban el número de anzuelos pescados en los cuadrados de 5° de latitud-longitud para 1958-2018. Cada flota tenía artes superficiales, intermedios o de profundidad. Estos artes simulados se asignaron a lances individuales basándose en patrones de pesca previamente comunicados o en la composición por especies de las capturas de la flota incluidas en la base de datos de ICCAT. Las profundidades de pesca de cada arte se adoptaron de un estudio anterior de la pesquería estadounidense. El número de lances en cada cuadrado se estimó a partir del total de anzuelos utilizando el número de anzuelos por lance para el arte. Los lances para cada pesquería en cada cuadrado de 5° fueron asignados aleatoriamente a celdas de LLSIM de 1° (sin tierra). El archivo resultante facilitaba en número de lances mensual para cada flota por arte dentro de cada cuadrado de 1° desde 1958 hasta 2018. Cuando se usa en combinación con los datos de entrada del modelo de distribución de

¹ 686 Hickory LN, Havana, Florida 32333, USA. phil_goodyear@msn.com

especies de aguja azul y pez espada, estas pesquerías permitirán a los investigadores simular capturas con ensayos de dos especies, tres pesquerías, del software LLSIM.

KEYWORDS

Longlines, CPUE, Stock assessment, Statistics, Simulation, Population modeling, LLSIM

1. Introduction

This paper documents the development of additional simulated longline fisheries suitable for inclusion in the longline simulator, LLSIM as recommended in Schirripa et al (2017). Data modeled after the ICCAT CPC longline fisheries for Japan, and Brazil were added to the existing effort data for the US used in earlier studies (Forrestal et al 2019a, 2019b). Information for the selected fisheries were extracted from the online version of ICCAT Task ii database downloaded 12/12/2020 and processed in a manner consistent with the methods used to construct the Japanese-like fishery in Goodyear et al (2018). These data provided the numbers hooks fished in 5° latitude-longitude squares for the period contained in the Task ii data. Each set in a 5° cell in the ICCAT data for the selected fisheries were randomly assigned to an included habitable 1° cell (no land). Previously-reported patterns in fishing depths were used to assign fishing gears for the Japanese-like fishery. For the Brazilian-like fishery, the gear used for each longline set was randomly assigned to shallow, intermediate, or deep gears based on the temporal patterns in the species composition of the catches. This task produced an LLSIM *effort* file that contain data for the synthetic fisheries suitable for LLSIM input, the software used to assemble this file, and the *gear* files referenced by the LLSIM *effort* file (see Goodyear 2018 for descriptions of the contents of LLSIM input files). The final assembly of files enable two-species, three-fishery runs of the LLSIM software.

2. Data for the individual ICCAT CPCs

2.1 Japanese-like fishery

Information about the Japanese fishery was extracted from the online version of ICCAT Task ii database. It was supplemented with information provided by the Japanese scientists in support of research for the ICCAT Methods Working Group (Goodyear 2006). These data were used to construct the Japanese-like synthetic fishery in Goodyear et al (2018). For the current study, the ICCAT data provide the numbers hooks fished in 5° latitude-longitude squares for the period 1956-2018. The additional information included 35 longline configurations deployed between 1956 and 1995 and the number of sets by gear type by year and month through 1995. A central feature of the data was a trend that increased the proportions of gears that fished deeper in the water column with time. However, the underlying detail changed markedly in the early 1970's. Goodyear (2006) showed a discontinuity in catchability computed with the data that seemed likely to be an artifact of data processing. To avoid the complicating consequences of these features, the simulated fishery patterned after the Japanese data was constrained to 3 generic gear types: shallow, intermediate, and deep. These replaced the 35 gears in the original files based on the relative depth fished by each gear. For 1956-1995 the gears in this simulated fleet were selected using the mean depths of the gears used in the 2006 study. Gear type after 1995 was randomly assigned based on the proportions in the 1995 data. The depth distributions of the simulated sets by year are in **Figure 1**. The shallow, intermediate, and deep fishing gears for the “Japanese” longlines were assigned to LLSIM gear files JAPshallow.csv, JAPintermediate.csv, and JAPdeep.csv, with gear codes of 203, 216, and 230, respectively, but the hook-fishing depths for these gear files were adopted directly from US gears 103, 116 and 130 with similar fishing depth assumptions. Since these files may not be representative of the actual fleet they should be carefully evaluated and possibly replaced in studies where inferences about the Japanese fishery are important. Each set in a 5° cell in the ICCAT data was randomly assigned to an included habitable 1° square (no land). For years prior to 1975 and after 1995, the number of sets in the 5° cells were estimated from the total number of hooks in the Task ii records using the average number of hooks per set by the gear for years where it had been used. There was a large temporal variability in the spatial distributions of sets by both month and year. This is illustrated by the contrasting distributions of the longline sets for 1975 in **Figure 2** and that for 2015 in **Figure 3**.

2.2 Brazil-like fishery

The Brazilian-like longline fishery was entirely developed using data extracted from the ICCAT Task ii database. Data representing the individual records were sorted based on the proportions of Swordfish and Yellowfin Tuna in the total catch of Yellowfin, Swordfish, Blue Marlin and Bigeye Tuna to assess probable fishing depth. For

both, the range increased from zero to a maximum proportion of 100% of the catch (**Figure 4**). The cumulative distribution of the proportion of yellowfin in the catch increased slowly near the origin but turned steeply near the end of the sampled distribution. In contrast, the proportion of Swordfish increased smoothly from zero to the maximum. There was no obvious discontinuity signifying different fishing methods. Consequently, the simulated fishery partitioned the gears used for each record in the ICCAT file into 3 equal groups. Records with proportions of Swordfish below the 33.3 percentile of the Swordfish proportion were assigned to a shallow gear, those between the 33.3 and 66.7 percentiles were assigned to the mid-depth gear, and those greater than that at the 66.7 percentile were assigned to the deeply-fishing gear. The result of this convention in proportions by year are shown in **Figure 5**. Given the shape of the cumulative for Yellowfin Tuna, it seems likely that this convention may overstate the relative proportions of shallow-fishing gears in the mix to be included in simulations. If so, this would diminish the probability that the simulated fishing effort is actually representative of the Brazilian fishery. The shallow, intermediate, and deep fishing gears for the Brazilian longlines were assigned to LLSIM gear files BRAShallow.csv, BRAintermediate.csv, and BRAdeep.csv, with gear codes of 402, 403, and 404, respectively. As for the Japanese-like fishery, the hook fishing depths for these gear files were adopted from US gears with similar fishing depth assumptions. Since these files may not be representative of the fleet they should be carefully evaluated and possibly replaced in studies where inferences about the Brazilian fishery are important. Each set in a 5° cell in the ICCAT task ii data was randomly assigned to an included habitable 1° cell. The resulting scatter of all sets from 1958 to 2018 is in **Figure 6**.

2.3 US-like fishery

Forrestal et al. (2019b) created a gear-effort matrix from longline logbooks that described the fishery and used Monte Carlo methods to create a data set describing a synthetic fishery resembling US longline fishing practices. These data covered the period 1986-2015 with 128 discrete gear types that were first used in a blind study of the performance of standardization methods (Forrestal et al 2019b). These data were adopted unchanged. The data file contained about 280 thousand simulated sets through 2015. The proportions of sets in deep, shallow and intermediate strata based on the fishing depths of each set in the data simulations is summarized in **Figure 7**. The 1986-2015 locations of sets are in **Figure 8**. Extending the data files backwards in time or through 2018 was beyond the scope of this study.

3. Assembly of the LLSIM input files

The final collection of files needed for an implementation of LLSIM using these new fisheries data must include species distributions, population trends, gears and set locations to be included in the simulation. The set locations are assembled in an “effort” file which concatenates the sets for each “fishery” for each month and each year from the beginning to the end of the time period simulated. The gear type is specified for each set. Each gear is defined in a separate file that specifies the gear identity to be used by LLSIM to index the gear. These files also identify the fleet and define separate species intrinsic catchabilities for each species to be included in a simulation (Goodyear 2018). The configuration for the gear specifies the number of hooks between floats and the hook depth probability distribution for each hook. One such sample collection of LLSIM input files was assembled. It utilizes the three fisheries described in **Section 2**. It also must employ species distribution data for Blue Marlin and Swordfish. The Blue Marlin data were based on the SDM described in Goodyear (2016) but implemented monthly with 1958-2012 environmental data. The Swordfish data were from the Swordfish SDM from Forrestal and Schirripa et al (2020). The Blue Marlin and Swordfish population trends were based on the most recent assessment results for each species (Anon. 2017, 2018) interpolated to 2018 (**Figures 9 and 10**). The annual estimates of the number of Swordfish in the catchable population was the sum of the assessment estimates for the North and South Atlantic. The South Atlantic values were computed from the estimated biomass divided by the mean weights in the North Atlantic. The time series for each species is included as one of the sample LLSIM input files that accompany this report. Each should be evaluated in the context of future investigations and replaced as necessary with trends consistent with the study objectives.

4. Discussion

Longline catch simulations using LLSIM need to evaluate the robustness of the simulations as a part of studies that utilize the simulated data. These evaluations will be influenced by features of the models used and the data employed. The sample gear files for the US fishery employed hook-depth probability distributions that were different for each hook position along the mainline. The fishing depths for the gear files for the “Japanese-like” and “Brazilian-like” fisheries were adopted from the US gears for convenience. In reality, for most gears the

proportions of time fished in shallower depth layers are higher nearer the floats. This feature is simulated in the LLSIM model and is reflected in the sample gear data developed for the USA fisheries (Forrestal et al 2019a, 2019b). Meanwhile, the species distributions varied by depth as well as latitude and longitude. Since competition for hooks influences the catch rate, changes in catch rates are not necessarily directly proportional to changes in abundance, and are influenced by the presence of other species. Such competitions may be important at some depths, not at others and may not be well represented by the means. This detail may or may not affect the utility of the simulated catches for studying principles needed for informative CPUE standardizations. For example, the simulated catches cannot mimic effects of variability in catch by hook position if the hook fishing depths in the gear files only reflect the means for each position between floats when different gears are pooled for convenience. The accuracy of this feature may or may not be important, depending on the intended application of the simulated CPUE data. This issue may be evaluated with LLSIM by partitioning a gear into multiple files with different hook depth probabilities. This example just illustrates the need to consider model performance characteristics as an important part of studies that implement LLSIM. The importance of competition for baits could be studied by adding a synthetic average fish “species” to represent nontarget catches.

The particular realizations of simulated catch data that are exhibited by variants of the datasets used in the sample run should be useful for many studies. However, they cannot represent the best assembly of data to test all specific questions about standardization practices. Particular issues may need datasets tailored to the situation. For example, a “fishery” with highly variable data by season can be constructed by assembling effort files that fish in the margins of the habitat (e.g., in the highest latitudes). Equatorial regions included in the oxygen minimum zone (OMZ, Prince et al 2010) would likely focus attention on vertical stratification. Other questions could employ different gears to analyze gear-related issues. Such studies can be addressed by assembling the needed input data from the available files or by creating new gear and fishing effort files as required. This in addition to improvement in the species distribution modeling.

Acknowledgements

This work was carried out under the provision of the ICCAT Science Envelope and the ICCAT – European Union Grant Agreement No. SI2.819116 - Strengthening the scientific basis for decision-making in ICCAT, and the ICCAT-US Data Fund. The Billfish Foundation supported the development of LLSIM. I thank M Schirripa for helpful comments and suggestions on all aspects of this work.

References

- Anon. 2017. Report of the 2017 ICCAT Atlantic Swordfish stock assessment session. Collect. Vol. Sci. Pap. ICCAT, 74(3): 841-967
- Anon. 2018. Report of the 2018 ICCAT Blue Marlin Stock assessment meeting. Collect. Vol. Sci. Pap. ICCAT, 75(5):813-888.
- Forrestal, F.C., and Schirripa, M. 2020. Addition of swordfish distribution model to longline simulator study. Collect. Vol. Sci. Pap. ICCAT, 77(5): 37-66 (2020).
- Forrestal, F.C., Schirripa, M., Goodyear, C.P., Arrizabalaga, H, Babcock, E., Coelho, R., Ingram., Lauretta, M., Ortiz, M., Sharma, R. and Walter, J. 2019a. Testing robustness of CPUE standardization and inclusion of environmental variables with simulated longline catch datasets. Fisheries Research 210: 1-13. <https://doi.org/10.1016/j.fishres.2018.09.025>
- Forrestal, F.C., Goodyear, C.P., Schirripa, M. 2019b. Applications of the longline simulator (LLSIM) using US pelagic longline logbook data and Atlantic Blue Marlin. Fisheries Research 211: 331–337.
- Goodyear, C.P. 2006. Performance diagnostics for the longline CPUE simulator. Collect. Vol. Sci. Pap. ICCAT, 59(2): 615-626.
- Goodyear, C.P. 2016. Modeling the time-varying density distribution of highly migratory species: Atlantic Blue Marlin as an example. Fisheries Research, 183: 469-481.

- Goodyear, C.P. 2018. Simulating Longline Catch with LLSIM: A User's Guide (Version 2). May 2018. <https://doi.org/10.13140/RG.2.2.11577.83040>.
- Goodyear, C. P., Schirripa, M. and Forrestal, F. 2017. Creating a species distribution model for Swordfish: evaluations of initial habitat variables. *Collect. Vol. Sci. Pap. ICCAT*, 74(3): 1235-1250 (2017).
- Goodyear, C. P., Schirripa, M., and Forrestal FC. 2017. Longline data simulation: a paradigm for improving cpue standardization *Collect. Vol. Sci. Pap. ICCAT*, 74(2): 379-390.
- Goodyear, C. P., Schirripa, M., Forrestal, F. and Laretta, M. 2018. Habitat covariates for standardizing longline cpue: an example with Blue Marlin. *Collect. Vol. Sci. Pap. ICCAT*, 75(5): 916-939 (2018)
- Prince, E. D., Luo, J., Goodyear, C. P., Hoolihan, J. P., Snodgrass, D., Orbesen, E. S., Serafy, J. E., et al. 2010. Ocean scale hypoxia-based habitat compression of Atlantic istiophorid billfishes. *Fish. Oceanogr.* 9, 448-462.
- Schirripa, M.J., Goodyear, C.P., and Forrestal, F. 2017. Longline data simulation: integrating 3-d species habitat with oceanographic data and depth distributions of pelagic longline hooks. *Collect. Vol. Sci. Pap. ICCAT*, 73(9): 3025-3034.

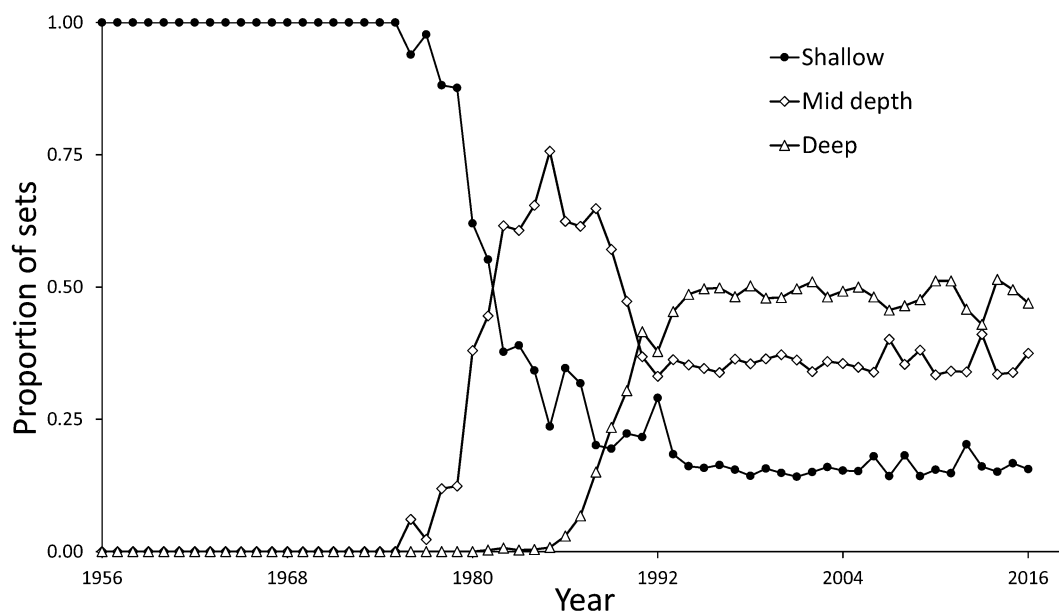


Figure 1. Proportions of shallow, intermediate and deep gears by year in the simulated fishery developed from ICCAT Task ii data using information about longline fishing depths reported by Japanese scientists.

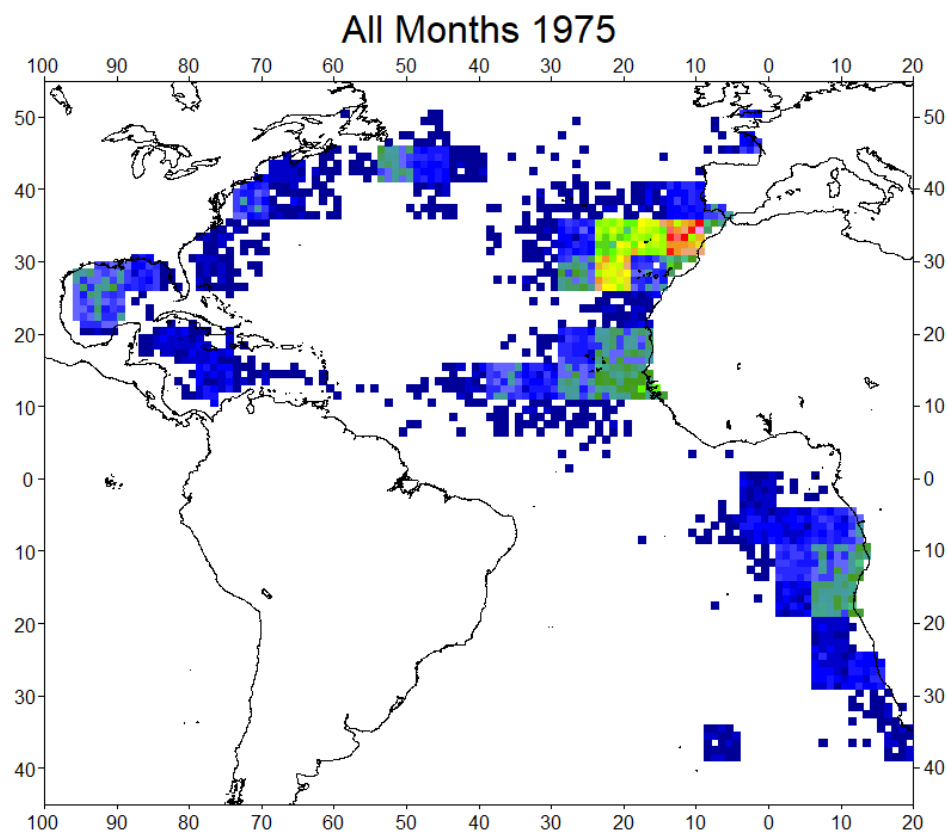


Figure 2. Distribution of sets in the 1975 simulated fishery based on ICCAT Task ii data for the Japanese longlines.

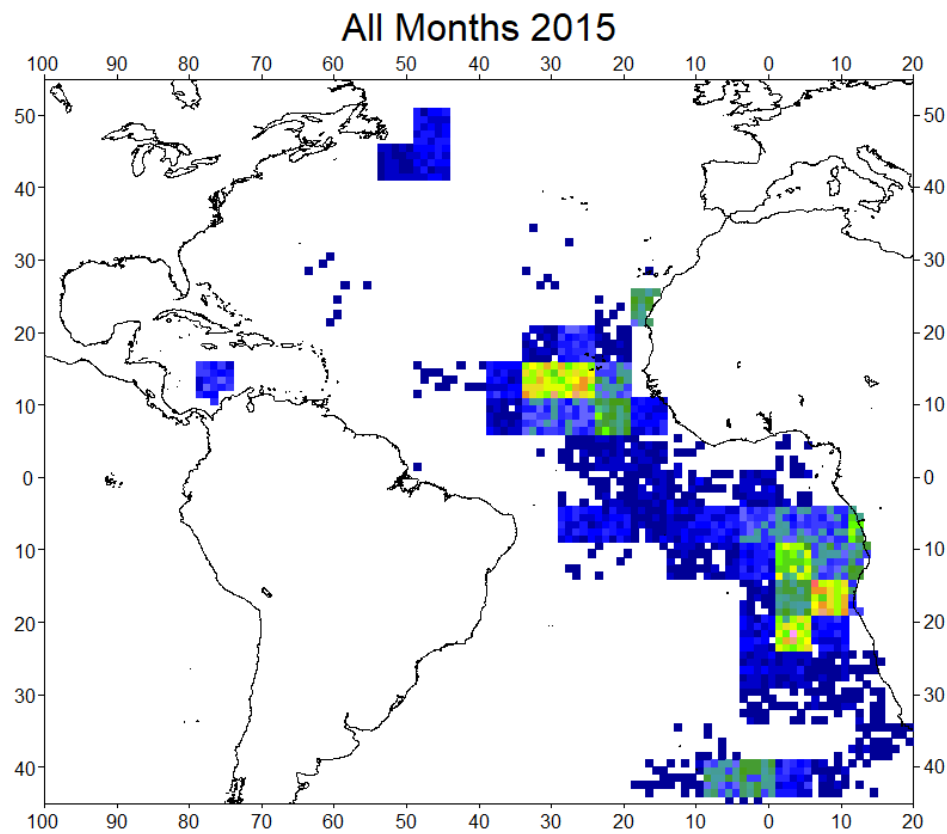


Figure 3. Distribution of sets in the 2015 simulated fishery based on ICCAT Task ii data for the Japanese longlines.

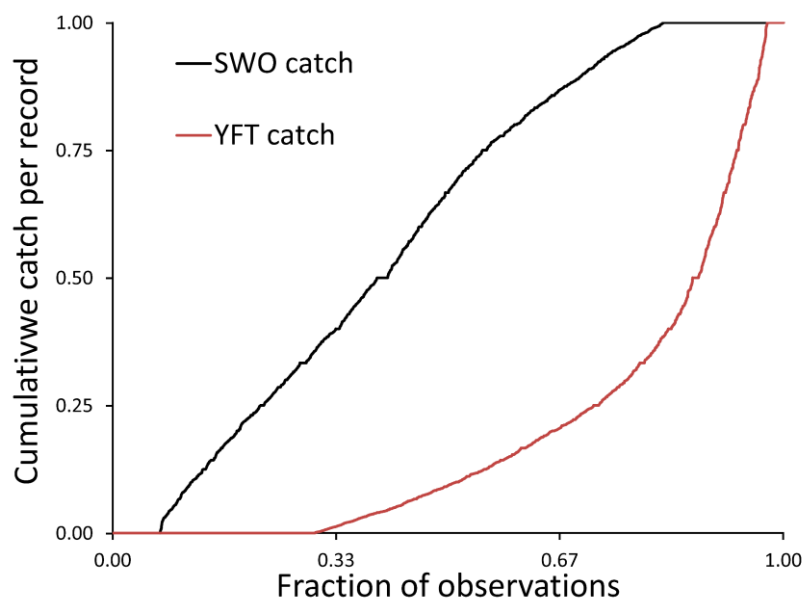


Figure 4. Cumulative fractions of total catch in each ICCAT data record for Swordfish and Yellowfin Tuna in the ICCAT Task ii data for the Brazilian longline fishery.

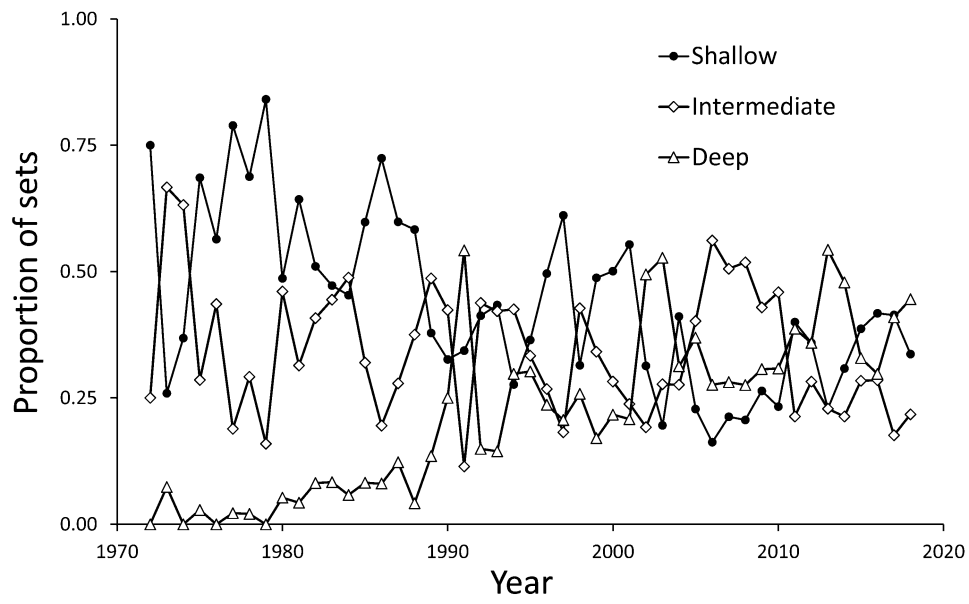


Figure 5. Proportions of shallow, intermediate and deep gears by year in the simulated fishery based on ICCAT Task ii data for the Brazilian longlines.

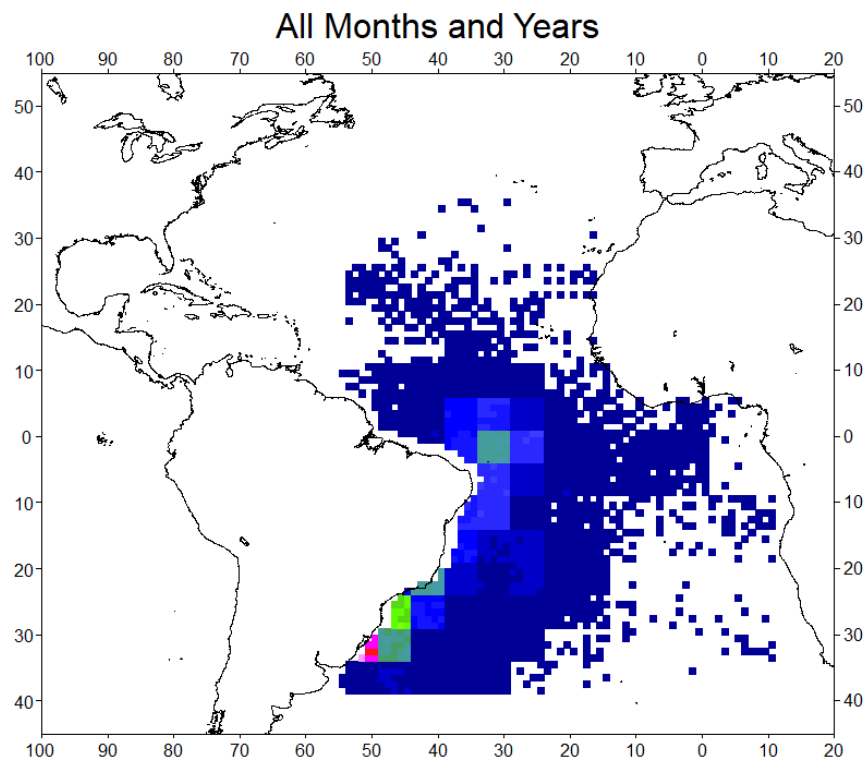


Figure 6. Distribution of sets in the simulated fishery based on ICCAT Task ii data for the Brazilian longlines.

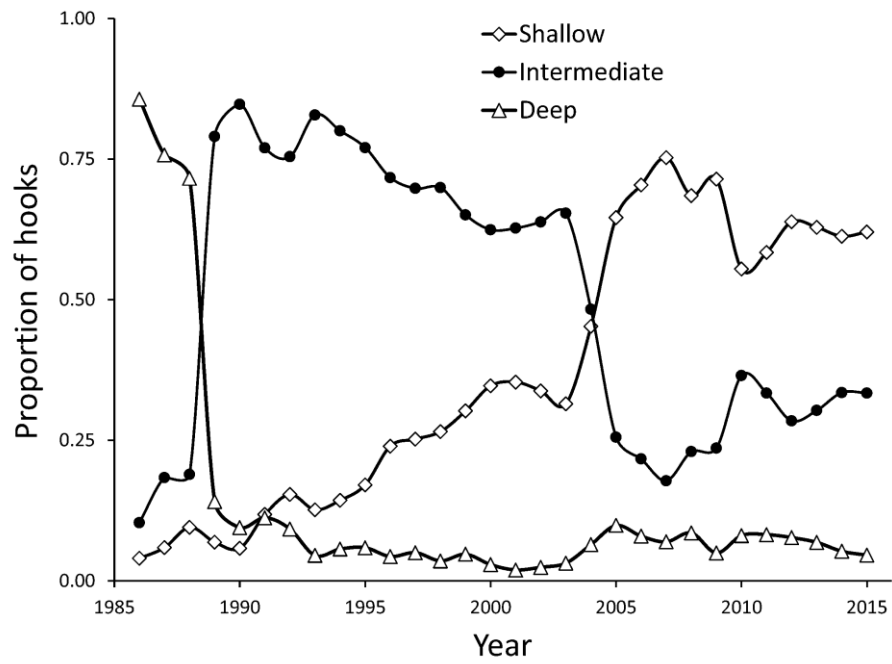


Figure 7. Proportions of shallow, intermediate and deep gears by year in the US Longline fishery developed from the longline simulations in Forrestal et al (2019).

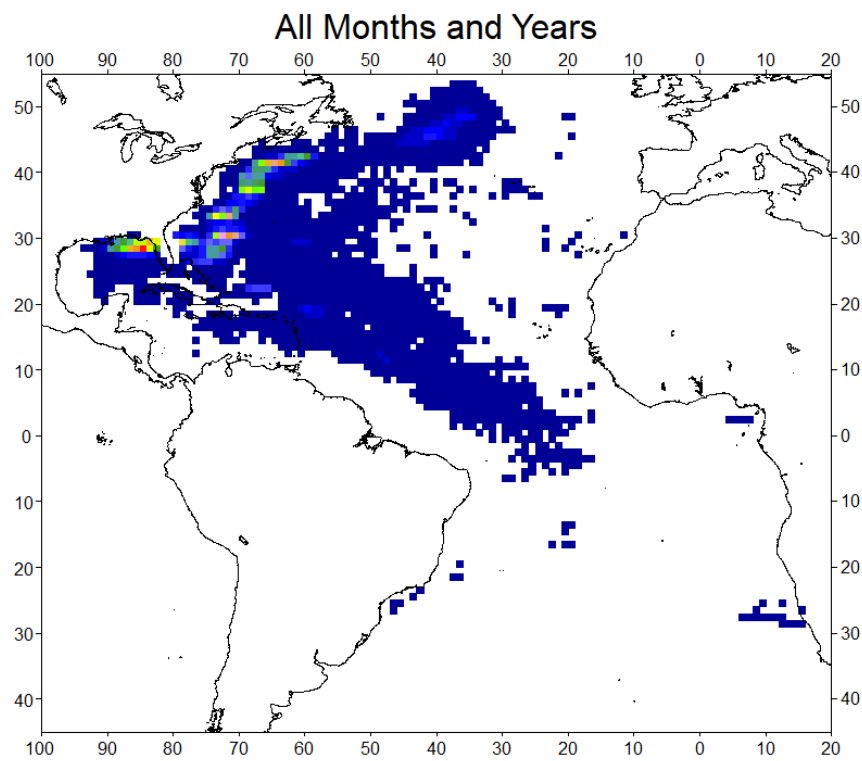


Figure 8. Distribution of sets in the 1986-2015 simulated fishery based on the USA longline logbook data.

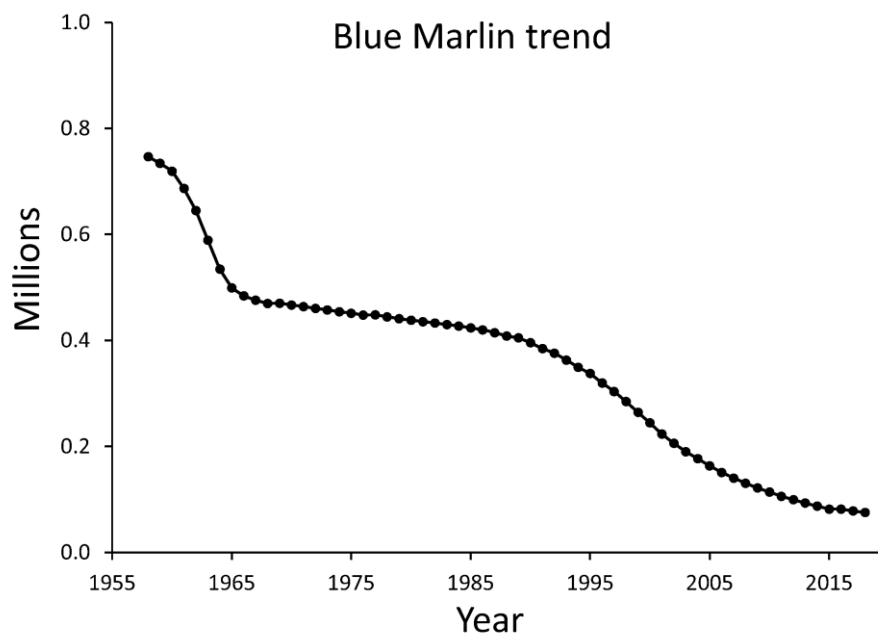


Figure 9. Annual abundances (millions of individuals in the catchable population) in the trend file for Blue Marlin used in the sample LLSIM run.

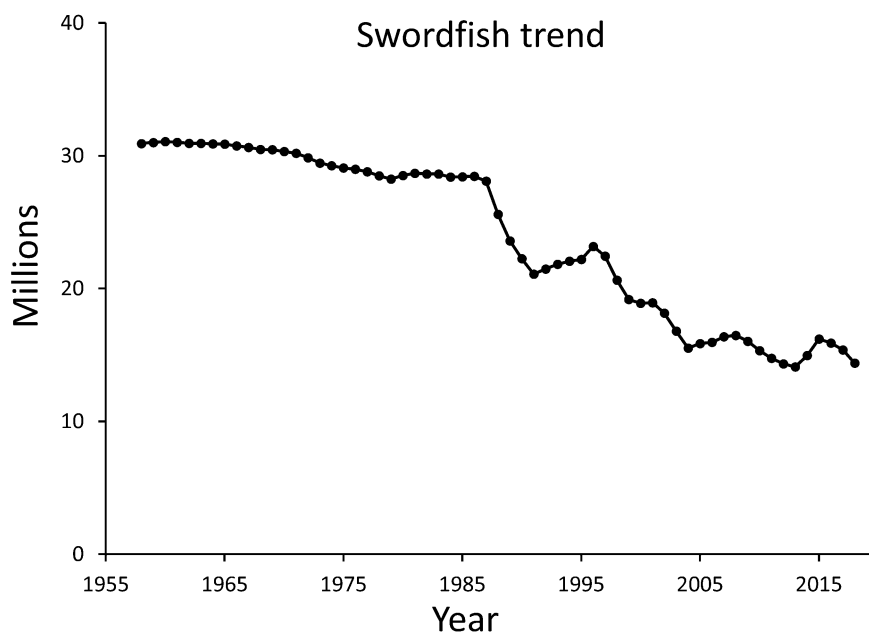


Figure 10. Annual abundances (millions of individuals in the catchable population) in the trend file for Swordfish used in the sample LLSIM run.